

Biometric Entrainment & The Vibe Audio Protocol (V.A.P.)

A Universal Standard for Experiential Audio Metadata

Abstract Current audio metadata standards (ID3) are insufficient for modern bio-adaptive applications. They capture bibliographic data (Artist, Title) but fail to capture experiential data (Affect, Biometrics, Context). This paper introduces the **Vibe Audio Protocol (V.A.P.)**, a 9-dimensional taxonomy that standardizes the storage of structural, emotional, and environmental audio data. We demonstrate how V.A.P. enables "Biometric Entrainment"—the synchronization of audio playback with human physiological states (Heart Rate, HRV) to optimize cognitive flow and physical performance.

1. The Problem: The "Metadata Gap"

In an era of smart devices and quantified self (wearables), music curation remains static. A 120 BPM track is treated identically whether it is a calming acoustic song or an aggressive industrial track. This "Metadata Gap" prevents algorithms from curating for *human state*.

2. The Solution: The 9-Pillar Taxonomy

V.A.P. expands audio definition into nine vectors:

1. **Structural:** Percussive DNA & Temporal Dynamics.
2. **Tonal:** Harmonic Complexity & Dissonance.
3. **Timbral:** Spectral Physics & Texture.
4. **Linguistic:** Semantic & Narrative Analysis.
5. **Affective:** Thayer Model Coordinates (Valence/Arousal).
6. **Contextual:** Scenario Probability Engines.
7. **Photometric:** Standardized Light/Color Mapping.
8. **Kinetic:** Biometric Entrainment Factors (MET Score).
9. **Genealogical:** Cultural Lineage & Authenticity.

3. Methodology: The Logic Architecture

V.A.P. utilizes a hybrid analysis model:

- **DSP Layer:** Extracts objective physical properties (Kick Transient, Spectral Centroid).
- **ML Layer:** Infers subjective properties (Mood, Scenario) using trained models.
- **GVS Layer:** Validates cultural properties (Tribe, Authenticity) via community voting.

4. Applications in "Scenario Computing"

We propose a "Contextual Override" system where playback logic shifts based on environmental inputs.

- *Case Study:* A user entering a gym triggers the **Kinetic Override**, rejecting tracks with a

MET Score < 6.0 regardless of user preference, enforcing a high-performance environment.

5. Conclusion

V.A.P. provides the necessary infrastructure for the next generation of "Empathic Computing," where media adapts to the physiological and emotional reality of the user.